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ADAN 8888 - Applied Analytics Project - Spring 2026
Module 3 Assignment - Perform Exploratory Data Analysis

HelpHerInvest: A machine learning-based investment recommendation system that aligns user interests and risk preferences with benchmark stock evaluation.

Dataset Partition Strategy

Our project consists of two components: an initial NLP-based filtering stage and a second supervised learning stage focused on performance prediction. The exploratory data analysis ("EDA") of the natural language processing ("NLP") component for this assignment is performed on the full dataset, since this stage is unsupervised and does not involve a labeled target variable. We will be utilizing this dataset to perform unsupervised semantic NLP, thus, a traditional train-test split is not applied during this phase. The discussion of dataset partitioning applies to the second part, which involves supervised learning on time-series financial data to develop a multinomial classification model based on the forward excess return.

To apply a time-aware partitioning strategy, the dataset is split chronologically by time period rather than randomly. The earliest 70% of the historical periods are used as the training set, where the model learns relationships between past financial features and forward excess returns. The next 15% of periods are reserved for validation and used for model tuning and intermediate performance checks. Then, the most recent 15% of periods will be used as a test set for final evaluation.

By splitting the data by time period, we avoid leaking future information into the training process and ensure the model performance is evaluated on truly unseen, forward-looking data. As a robustness check, we may also compare results against a randomly shuffled split in later analysis to examine how model performance differs.

Semantic Natural Language Processing

EDA Analysis and Insights

Exploratory data analysis was conducted on the full dataset used in the NLP-based similarity component to evaluate whether the available text fields support the interest alignment step described in Week 1. The analysis focused on data coverage, categorical structure, text length

characteristics, and the presence of duplicated or boilerplate language, with supporting figures and tables provided in **Appendix A**.

The EDA shows that the dataset supports a two-stage recommendation system, but primarily for instrument types with complete and reliable metadata. Sector and industry information is missing for 1,812 out of 10,284 rows (17.6%), while the business summary field is missing for 1,103 rows (10.7%). Missingness is not uniform across instrument types: ETFs and mutual funds exhibit near-complete missingness for sector and industry fields, while equities show substantially lower missing rates (**Figure A1**). This suggests that the NLP-based interest alignment step is feasible mainly for equities and would be unreliable for ETFs and mutual funds unless external metadata is incorporated.

Sector and industry labels exist at a meaningful scale, with 11 unique sectors and 145 unique industries, but the distribution is highly unbalanced (**Figure A2**). While this allows coverage across many economic categories, imbalance implies that similarity-based retrieval may disproportionately surface stocks from larger sectors unless normalization or diversification constraints are applied. This has direct implications for user experience, as recommendations may otherwise lack breadth across industries.

Analysis of the business summary text indicates that descriptions are generally long enough to support semantic similarity modeling. For non-empty summaries, the average length is approximately 133 words, with a maximum of 746 words (**Figure A3**). However, the text contains heavy boilerplate language and a large number of exact and near-duplicate summaries (**Figure A4**). In total, 2,597 rows belong to duplicated summaries, with some descriptions repeated up to 22 times (**Table A5**). These patterns can dilute discriminative signals, cause repetitive recommendations, and artificially inflate evaluation metrics if duplicated text appears across training and validation splits.

Data Problems Identified

Based on the EDA findings, several data quality issues were identified that directly impact model reliability and evaluation. First, uneven coverage of NLP fields by instrument type suggests the need to either filter the NLP pipeline to equities or augment metadata for other instruments. Second, duplicated and near-duplicated business summaries reduce effective data diversity and increase the risk of leakage during evaluation. Third, boilerplate-dominated text may overshadow user-relevant themes, while extreme text lengths and very short summaries create instability for embedding-based methods.

Preprocessing Recommendations

To address the data problems stated above, we recommend the following preprocessing steps prior to NLP:

1. Limit the NLP pipeline to equities, at least initially. The EDA shows that sector, industry, and business summary fields are frequently missing for ETFs and mutual funds, which makes similarity scoring unreliable for those instruments unless additional metadata is added.
2. Address duplicated and near-duplicated summaries before modeling. Many companies share identical or nearly identical descriptions, so these should be collapsed into a single representation. Duplicated text should also be kept within the same evaluation split to avoid inflated similarity results.
3. Reduce the impact of boilerplate language in company descriptions. Common phrases that appear across many summaries can dominate scores, so basic text cleaning such as lowercasing, punctuation removal, and stopward filtering should be applied, with the option to remove especially repetitive phrases.
4. Deal with text length variation and missing summaries in a consistent way. Very long summaries should be truncated based on the observed length distribution, while missing summaries should be handled explicitly rather than dropped. Extremely short summaries may be excluded from NLP matching or flagged as low-confidence cases.

Multinomial Classification Model

EDA Analysis and Insights

The EDA technical analysis financial dataset provided us more insight on the features and target variable (forward excess return, or “fwd_excess”). See **Appendix B** for more details.

Due to how the values are calculated based on historical data, there are several null values for certain columns that rely on lagging data, such as 6-month momentum, 12-month momentum, 6-month drawdown, 12-month drawdown, etc. **(Figure B1)**. We will have to accommodate for this by either manipulating the data or adjusting the time frame in which we are pulling historical data from.

By sorting the average “fwd_excess” values by the tickers, you can see that some of the most popular stock symbols have provided the biggest return. Symbols such as TSLA, NFLX, NVDA, META, or AMZN are present in the top 10, which is not overly surprising. (**Figure B3**)

Many of the features are normally distributed, but some features are skewed. Specifically the volatility variables are slightly right-skewed and the drawdown variables are slightly left-skewed. **(Figure B5)**.

The correlation analysis notes that there are no strong positive or negative correlations between the numeric features and fwd_excess. A straightforward model based on a few linear signals will likely underperform and may need non linear learners interactions regime awareness or alternative targets such as classification into quartiles as planned in Week 1. **(Figure B6)**

Potential Problems Identified

1. The number of null values for some features could be problematic and produce inaccurate results.
2. In the supervised target “fwd_excess”, the distribution has a tail right (rightly skewed) which can indicate outliers. The max value is 4.02 in the training set, where the mean value is 0.11. Outliers can dominate loss functions and can cause the model to overfit to a small number of extreme events **(Figure B5)**.
3. Some features show skewness that may affect modeling assumptions. This can impact models sensitive to distribution shape and can impact prescription results **(Figure B5)**
4. Weak global linear relationships between features and target correlation analysis concludes there are no strong positive or negative correlations with fwd_excess. This is not a data quality issue by itself, but it indicates the supervised learning problem may require non linear models, interaction effects, time varying relationships, or additional features **(Figure B6).**

Preprocessing Recommendations

Create an explicit missing data strategy for numeric features with systematic missingness. We can apply the following methods across the train validation test:

1. Restrict the supervised dataset to rows where all required lookback features exist, which simplifies modeling but reduces coverage and may exclude newer tickers. Adjustments to date range may make this possible.

2. Impute missing values using time aware approaches such as per ticker rolling median or cross sectional median by date and add binary missingness indicators so the model can learn that the value was missing.
3. Use models that can handle missing values directly where appropriate and still consider adding missingness flags Also recommended is to document the cause of missingness as insufficient history and treat it as an expected structural condition rather than random noise.

Handle outliers in “fwd_excess” and consider robust targets due to the rightly skewed results. Research applying one or more of the following preprocessing steps for supervised learning:

1. Winsorize or clip fwd_excess at the 1st and 99th percentiles and compare model stability to the unclipped target
2. Use robust losses or switch to the planned classification framing in Week 1 such as quartiles for ranking stocks, which reduces sensitivity to rare extreme values
3. Also consider per date normalization of the target if your goal is cross sectional outperformance rather than absolute returns

Dealing with the features which are not normally distributed.

1. Utilize robust scaling methods such as median and IQR scaling and consider transforms for strictly positive skewed measures if applicable
2. Potentially utilize standardization methods across the features to normalize the distribution.

Determine the dataset partition strategy used.

1. The current split maintains a time series ordered 70/15/15 split on approximately 9,200 rows. Ensure rows are truly ordered by date and that no future information is embedded in features relative to the prediction horizon. Using time series splits rather than random splits aims to preserve realistic deployment conditions and potential autoregressive nature of values.
2. Random split approach could perform better and eliminate autoregressive bias from the predictions.

Appendix A

NLP Exploratory Data Analysis

Figure A1: Missingness Rate by quoteType (Sector/Industry/Summary)

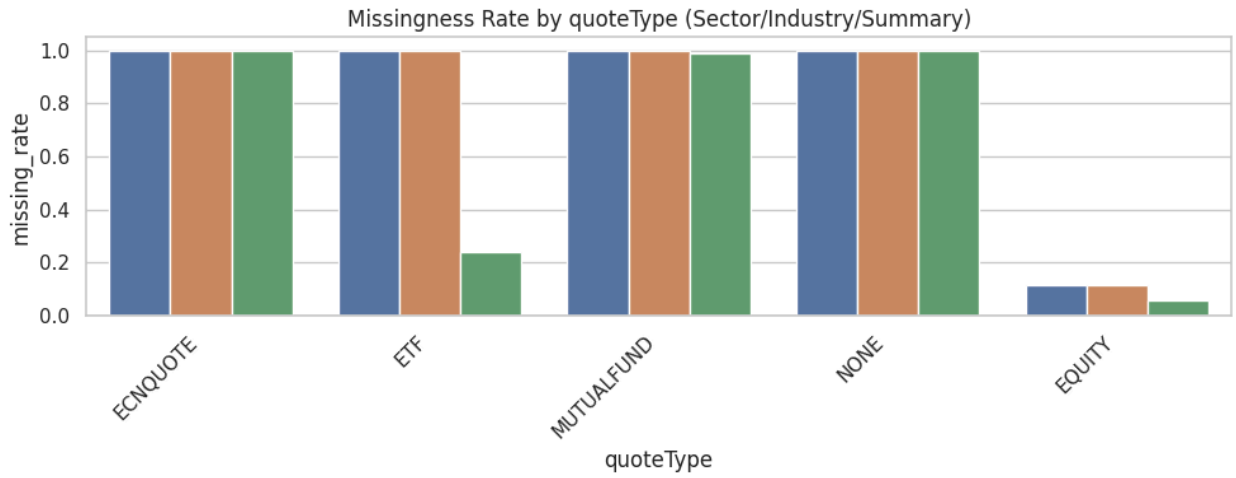


Figure A2: Sector Distribution Across Equity Instruments

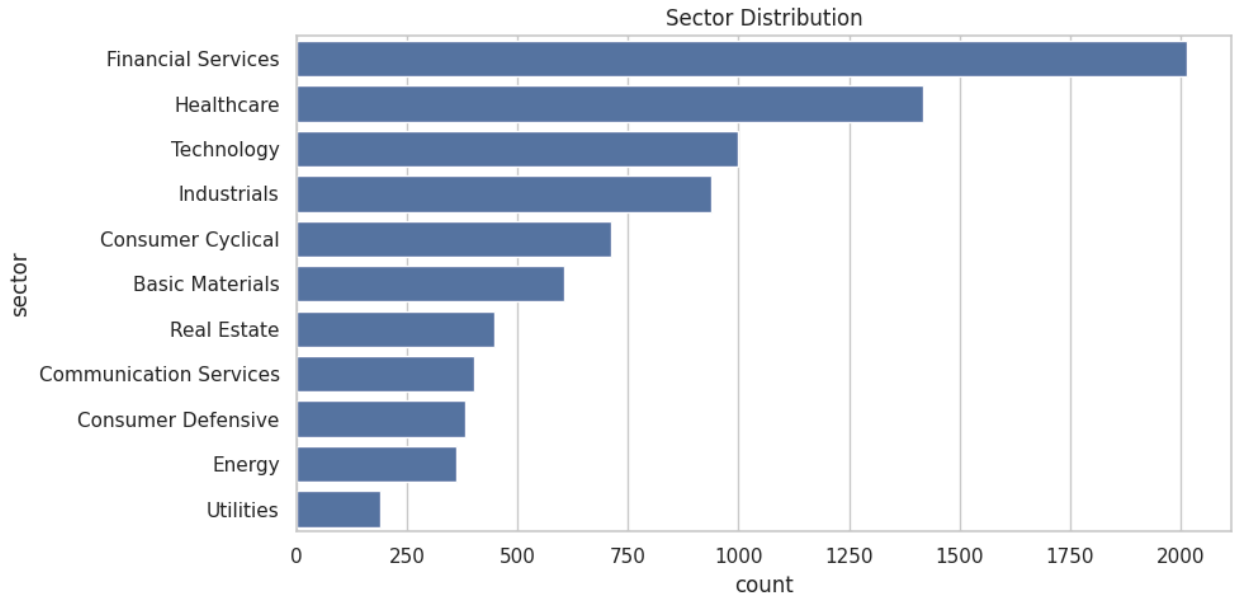


Figure A3: Business Summary Character Count (Clipped at 99th percentile)

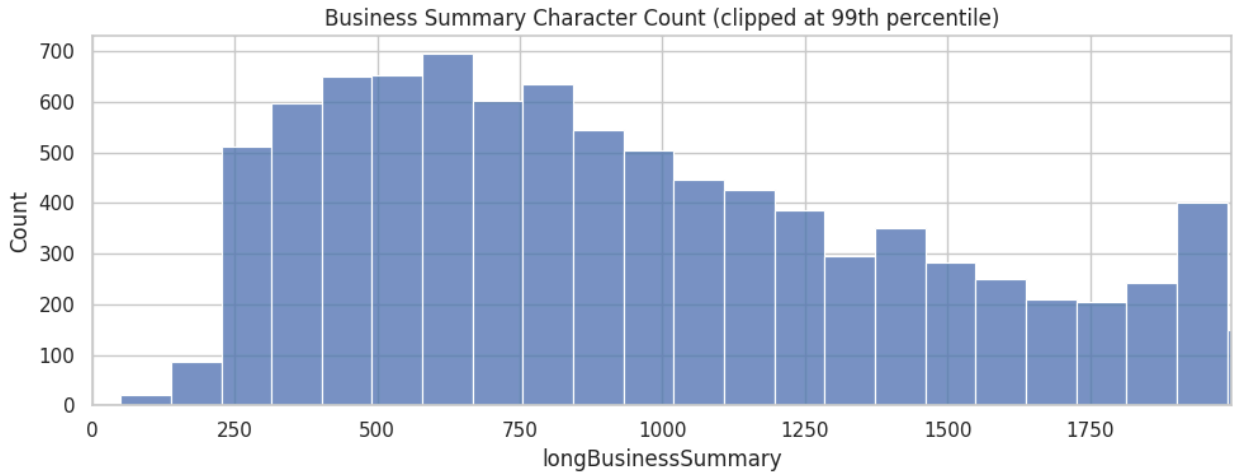


Figure A4: Top Unigrams and Bigrams in Cleaned Business Summaries

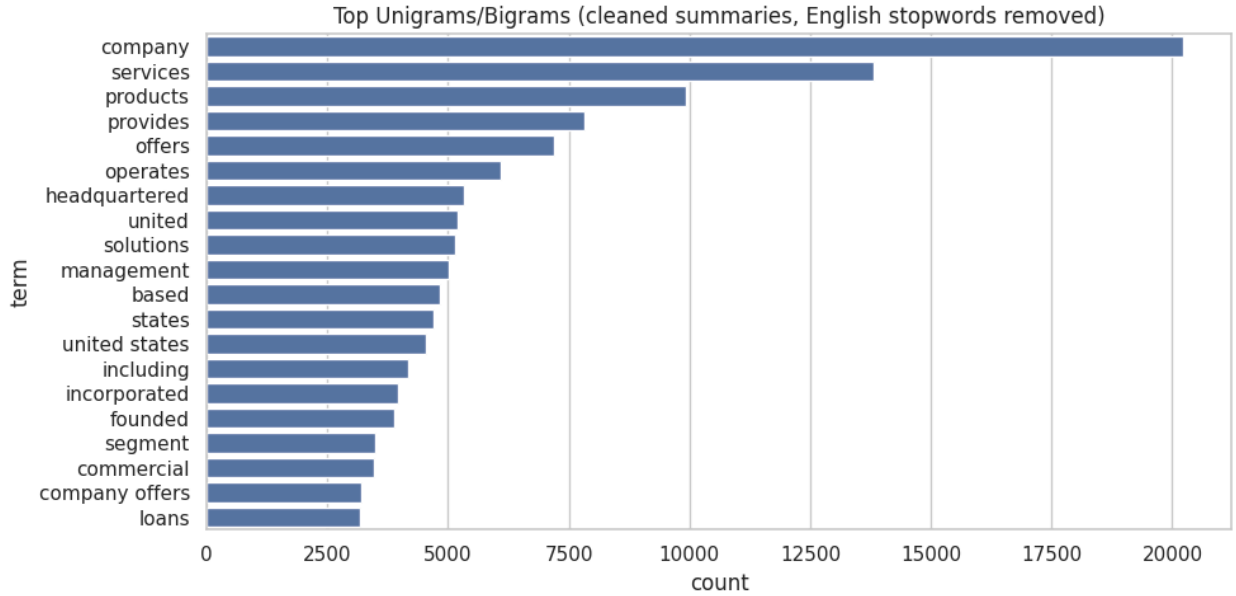


Table A5: Summary Statistics of Duplicate Business Summaries

	distinct_nonempty_summaries	distinct_duplicated_summaries	rows_in_duplicated_summaries	max_repetition_single_summary
0	7628	1044	2597	22

Appendix B

Classification Model Exploratory Data Analysis

Figure B1: Missing Values per Column

```
Missing values per column:
Date                  0
Ticker                0
mom_1m                0
mom_3m               96
mom_6m              240
mom_12m             526
mom_12m_ex_1m       526
rel_3m_spy           96
rel_6m_spy          240
rel_12m_spy         526
vol_3m               96
vol_6m              241
drawdown_6m         192
drawdown_12m        479
pct_above_200dma    388
adj_close             0
fwd_excess            0
fwd_return            0
```

Figure B2: Descriptive Summary Statistics

	fwd_excess	mom_1m	mom_6m	mom_12m	adj_close
count	6312.000000	6312.000000	6072.000000	5786.000000	6312.000000
mean	0.034056	0.002863	0.005591	0.008231	63.114141
std	0.233581	1.006955	1.008082	1.008781	60.450825
min	-0.742708	-4.941975	-4.855368	-3.909592	0.210643
25%	-0.083384	-0.568056	-0.569720	-0.529152	22.894295
50%	0.011648	-0.047140	-0.105291	-0.147831	45.524954
75%	0.111277	0.522415	0.454105	0.346506	83.734716
max	4.020147	6.164418	6.533410	6.526852	662.283752

Figure B3: Mean Forward Excess Sorted by Ticker

	count	mean	std	min	25%	50%	75%	max
Ticker								
TSLA	132.0	0.343586	0.800536	-0.511812	-0.111229	0.086031	0.365226	4.020147
NFLX	137.0	0.203107	0.501023	-0.698185	-0.106404	0.151981	0.385885	2.376051
NVDA	137.0	0.195739	0.367165	-0.436245	-0.029262	0.085357	0.415260	1.423459
AMD	137.0	0.147773	0.502850	-0.742708	-0.204725	0.041220	0.410807	2.323218
AVGO	137.0	0.105272	0.165957	-0.196993	-0.028021	0.086841	0.197614	0.566537
META	109.0	0.101172	0.218840	-0.293504	-0.025067	0.062503	0.184757	1.033359
AMZN	137.0	0.095740	0.205740	-0.321286	-0.041970	0.070517	0.245249	0.640745
ASML	137.0	0.093080	0.146323	-0.213727	0.021243	0.095200	0.168726	0.513613
MU	137.0	0.091248	0.348412	-0.472740	-0.212027	0.052406	0.326857	1.122790
AAPL	137.0	0.084553	0.189100	-0.430331	-0.043496	0.103470	0.213669	0.705514
LRCX	136.0	0.082489	0.189970	-0.358800	-0.043723	0.073878	0.212969	0.610807
UNH	137.0	0.061534	0.104059	-0.199528	-0.002127	0.055271	0.122415	0.378020
MA	137.0	0.061468	0.123499	-0.227333	-0.018367	0.060947	0.135700	0.445558
HD	137.0	0.056975	0.106610	-0.181832	-0.029390	0.055351	0.133885	0.340476
MSFT	137.0	0.055061	0.112606	-0.178858	-0.025678	0.071170	0.129963	0.329686
V	137.0	0.043824	0.118697	-0.236447	-0.022832	0.060026	0.127745	0.306520
GOOGL	137.0	0.043584	0.119453	-0.275899	-0.032913	0.031590	0.107088	0.439198

Figure B4: Aggregated Mean Forward Excess

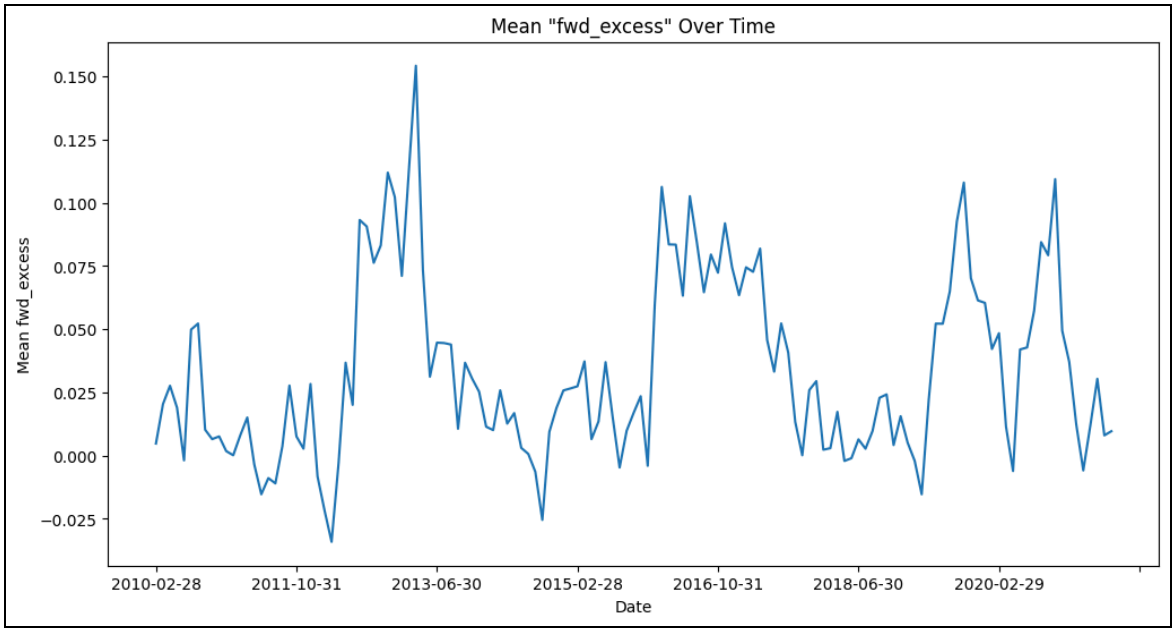


Figure B5: Histograms of Variable Distributions

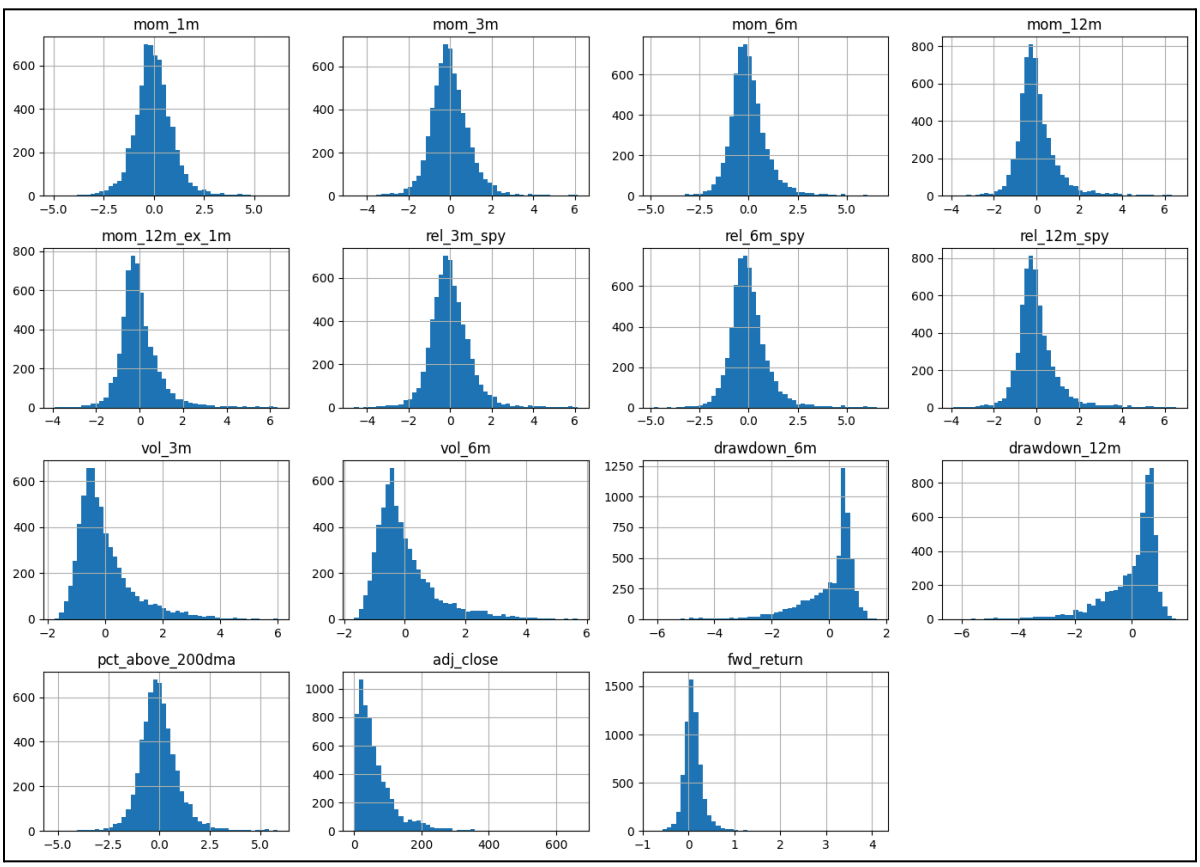
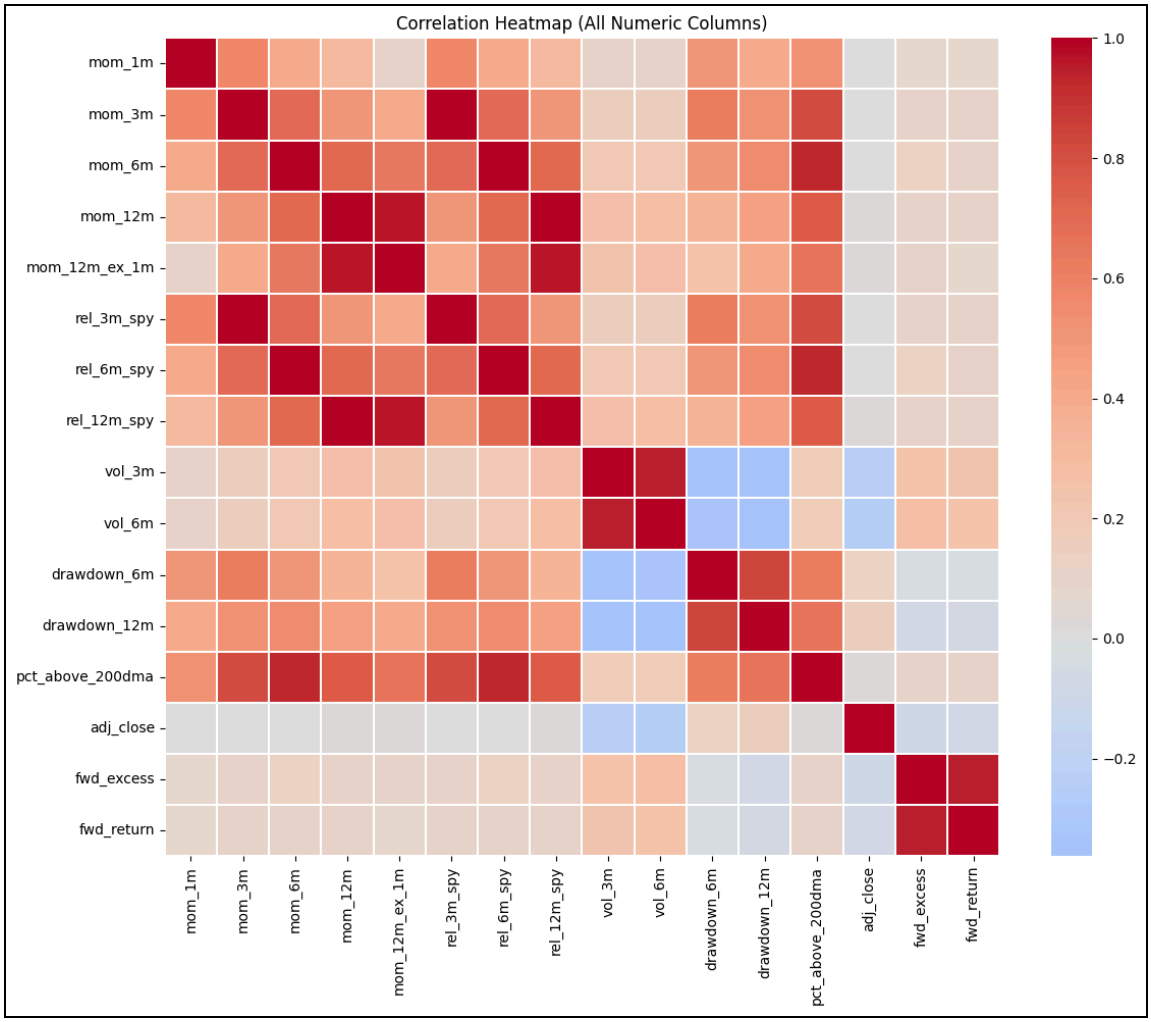


Figure B6: Correlation Heatmap for Variables


Correlation Heatmap (All Numeric Columns)
mom_1m
mom_3m
mom_6m
mom_12m
mom_12m_ex_1m
rel_3m_spy
rel_6m_spy
rel_12m_spy
vol_3m
vol_6m
drawdown_6m
drawdown_12m
pct_above_200dma
adj_close
fwd_excess
fwd_return
1.0
0.8
0.6
0.4
0.2
0.0
−0.2